ChargeAtlas India

Chat History — Last 12 Prompts & Responses

Project: EV Infrastructure Map India | IIT Bombay Techfest 2026-27 | Team SuperNova
Generated: 04 November 2026, 12:17 IST

Prompt 1 of 12 2026-11-02T09:35:59Z

USER

complete this work proeprly first

ANTIGRAVITY (Agent)

[No response recorded]

Prompt 2 of 12 2026-11-02T09:57:26Z

USER

CHARGEATLAS INDIA â€" AUTHENTIC DATA ACQUISITION, VALIDATION AND PRODUCTION INGESTION

Project directory: D:\EV Infra

The PostgreSQL and PostGIS foundation is ready.

Now establish the authentic data foundation for ChargeAtlas India.

PRIMARY OBJECTIVE

Replace synthetic data in production analytical views with authentic, traceable and reproducible data from official sources.

The application must never present synthetic or unverified information as official statistics.

1. DATA SOURCE DISCOVERY

Investigate these sources.

A. EV registration data

Official source:
https://analytics.parivahan.gov.in/analytics/publicdashboard/vahan?lang=en

Required:
- Maharashtra EV registration totals.
- District/RTO-level records where accessible.
- 2W registrations.
- 3W registrations.
- 4W registrations.
- Electric buses.
- Historical year-wise or month-wise data where available.
- Registration period and extraction date.

Do not assume that the dashboard exposes a downloadable API.

Use official downloadable exports or permitted access methods.

If an export requires manual interaction, provide exact instructions for obtaining it.

B. Charging station data

Official source:
https://evyatra.beeindia.gov.in/

Required:
- Charging station name.
- Latitude and longitude.
- Address.
- Operator.
- Station status, where available.
- Access classification.
- Charging point information.
- Connector information.
- Rated power.
- Source update date.

Investigate whether official exports or documented APIs are available.

C. Geographic boundaries

Investigate:
- Survey of India.
- Official Maharashtra government GIS sources.
- Local municipal planning sources.

Required:
- Maharashtra state boundary.
- District boundaries.
- Pune and PCMC urban boundaries, with clear definition.
- Other geographic boundaries required for the study.

Preserve source dates and administrative changes.

Do not invent missing district geometries.

D. Samruddhi Mahamarg

Official source:
https://msrdc.in/

Required:
- Route alignment.
- Interchange coordinates.
- Operational/planned interchange classification.
- Wayside facilities.
- Carriageway access information, if available.

Do not infer official highway geometry from an approximate line if official alignment data is

available.

E. Population

Official source:
https://censusindia.gov.in/

Required:
- District population.
- Census reference year.
- Relevant demographic/geographic data.
- Any projection methodology used.

Do not label projections as census counts.

F. Electricity infrastructure

Investigate official MSEDCL and MSETCL sources.

Required where accessible:
- Substation coordinates.
- Voltage level.
- Feeder information.
- Spare capacity.
- Feeder loading.
- Source date.

Geographic proximity alone must not be described as proof of electrical feasibility.

G. Charging policy

Use the official Ministry

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ANTIGRAVITY (Agent)

[No response recorded]

Prompt 3 of 12 2026-11-02T10:16:29Z

USER

continue the work properly

ANTIGRAVITY (Agent)

[No response recorded]

Prompt 4 of 12 2026-11-02T10:26:45Z

USER

CHARGEATLAS INDIA â€" COMPLETE GIS ENGINE AND EV INFRASTRUCTURE GAP ANALYSIS

Project: D:\EV Infra

Use the existing PostgreSQL/PostGIS database and the authentic datasets that have passed the previous data-validation stage.

OBJECTIVE

Complete the geographical intelligence engine of ChargeAtlas India.

The system must calculate actual infrastructure coverage, identify gaps, and provide traceable results.

1. INDIA AND MAHARASHTRA GEOGRAPHIC ENGINE

Implement:
- State boundary analysis.
- District boundary analysis.
- Point-in-polygon station assignment.
- District-level EV registrations.
- District-level station counts.
- Charging-point distribution.
- Connector distribution where available.
- EV-to-charging-point ratio.
- EV-to-station ratio.
- Charging infrastructure density.

Clearly distinguish unavailable data from a true zero.

2. SPATIAL VALIDATION

Use PostGIS.

Implement and test:
- ST_Within.
- ST_Intersects.
- ST_DWithin.
- ST_Distance.
- ST_Transform.
- ST_Area.
- ST_Length.

Use appropriate geography or projected CRS for distance and area calculations.

Do not assume one UTM zone is suitable for all of Maharashtra or the full Samruddhi corridor.

Store original geometries in EPSG:4326.

Use EPSG:3857 only where appropriate for map display.

3. DISTRICT ANALYSIS

For each district, calculate where data permits:

- EV registrations by vehicle category.
- Total physical charging stations.
- Total charging points.
- Total connectors.
- EV-to-charging-point ratio.
- Population-adjusted infrastructure indicators.
- Spatial coverage.
- Infrastructure deficit indicators.

Show:
- District ranking.
- District comparison.
- Data availability.
- Source and reporting year.

Do not compare inconsistent reporting periods without disclosing the difference.

4. PUNE URBAN GRID ANALYSIS

Use the verified Pune urban boundary.

Generate a 1 km Ã— 1 km grid using an appropriate metric CRS.

Clip grid cells to the study boundary.

Calculate:
- Grid-cell area.
- Number of stations within each cell.
- Number of cells covered.
- Number of cells without a station.
- Distance to nearest station.
- Coverage percentage.
- Deficit percentage.

Use the 2024 Ministry of Power urban planning benchmark:

At least one charging station within a 1 km Ã— 1 km grid in urban areas by FY 2029â€“30, subject to the applicable notified urban-area definition.

Do not automatically classify every Pune grid cell as legally non-compliant unless the relevant planning geography and policy applicability are established.

Distinguish a planning benchmark from actual legal compliance.

5. SAMRUDDHI MAHAMARG ANALYSIS

Use verified route alignment and interchange records.

Implement:
- Route length.
- Interchange locations.
- Route segmentation.
- Distance between facilities.
- Nearby charging stations.
- Corridor coverage.
- Identified gaps.

Use the Ministr

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ANTIGRAVITY (Agent)

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Prompt 5 of 12 2026-11-02T11:09:41Z

USER

CHARGEATLAS INDIA â€" EV DEMAND FORECASTING AND INTELLIGENT CHARGING SITE RECOMMENDATION ENGINE

Project directory: D:\EV Infra

Use the existing authentic PostgreSQL/PostGIS data and validated GIS analytics.

Do not build a separate application.

OBJECTIVE

Develop an explainable forecasting and candidate charging-site recommendation system.

The output must support infrastructure planning, not pretend to provide guaranteed future outcomes or engineering approvals.

PART A â€" EV DEMAND FORECASTING

1. Historical baseline

Use authentic historical VAHAN records.

Separate:
- 2W.
- 3W.
- 4W.
- Buses.

Record:
- Baseline year.
- Reporting period.
- Registration count.
- Source.
- Data completeness.
- Geographic coverage.

If historical observations are insufficient, clearly identify that limitation.

Do not invent historical registrations.

2. Forecast models

Implement appropriate methods depending on data availability.

Consider:
- CAGR scenario modelling.
- Linear trend.
- Exponential growth.
- Time-series methods where sufficient historical observations exist.

Do not claim a sophisticated machine-learning model is accurate simply because it was implemented.

Compare models using suitable historical validation where possible.

3. Forecast years

Generate planning scenarios for:
- 2027.
- 2030.

Provide:
- Low-growth scenario.
- Base scenario.
- High-growth scenario.

Use published, defensible assumptions where available.

Where assumptions are not sourced, make them configurable and label the results hypothetical.

Do not present arbitrary CAGR values as official projections.

4. Charging demand estimation

Estimate:
- Number of EVs.
- Daily energy requirement.
- Peak charging demand.
- Required charging points.
- Charging capacity by vehicle segment.
- Infrastructure deficit.

Document all equations and input assumptions.

Include:
- Daily kilometres.
- Energy consumption per kilometre.
- Charging efficiency.
- Charging behaviour.
- Diversity and utilization assumptions.

- Charging-point power.
- Operating hours.

Do not use one energy consumption constant for every vehicle category.

Where reliable inputs are unavailable, show a scenario range.

5. Forecast interface

Provide:
- Year selector.
- Vehicle-category filters.
- Growth scenario selector.
- Custom growth assumption controls.
- Forecast charts.
- Historical versus projected comparison.
- Charging demand estimates.
- Assumption inspector.
- Data-source references.

Display uncertainty and scenario labels.

PART B â€" CANDIDATE CHARGING SITE RECOMMENDER

6. Candidate locations

Use real, geospatially valid candidate locations derived from:
- Identified infrastructure gaps.
- Road network accessibility.
- EV demand.
- Existing station coverage.
- Verified public land or candidate parcels where accessible.
- Electricity infrastructure.

Do n

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ANTIGRAVITY (Agent)

[No response recorded]

Prompt 6 of 12 2026-11-02T17:44:04Z

USER

CHARGEATLAS INDIA â€" FINAL PROFESSIONAL GIS FRONTEND AND USER EXPERIENCE IMPLEMENTATION

Project: D:\EV Infra

Improve the existing React application into a professional GIS-based infrastructure intelligence platform.

Do not create a new frontend.

Preserve the working architecture, existing business logic and API contracts.

1. DESIGN DIRECTION

Design ChargeAtlas as a serious geospatial decision-support application.

Visual requirements:
- Professional.
- Minimal.
- Modern.
- Clean.
- Information-rich.
- High readability.
- Consistent.
- Responsive.

Avoid:
- Neon colours.
- Generic AI aesthetics.
- Excessive gradients.
- Glassmorphism everywhere.
- Decorative animations.
- Oversized cards.
- Unnecessary icons.
- Excessive rounded corners.
- Flashy dashboard effects.

Use a restrained colour system with meaningful semantic colours.

Maintain dark mode and a well-designed light mode.

2. APPLICATION SHELL

Improve:
- Navigation sidebar.
- Header.
- Breadcrumbs.
- Search.
- Data-source status.
- Database connection status.
- Notifications.
- User account area where authentication exists.

Navigation must remain consistent across all modules.

Provide a clean responsive sidebar on desktop and an appropriate mobile navigation layout.

3. OVERVIEW DASHBOARD

Display authentic metrics only in authentic mode.

Show:
- EV registration totals.
- Charging stations.
- Charging points.
- Charging connectors.
- EV-to-charger ratios.
- District-level distribution.
- Vehicle category composition.
- Infrastructure coverage.
- Data freshness.
- Data-quality indicators.

Every metric must disclose its:
- Source.
- Reporting period.
- Verification status.

Provide meaningful charts, not decorative charts.

Do not hardcode old metrics.

4. NATIONAL MAP

Implement:
- India-wide map.
- State boundaries.
- Station clustering.
- State selection.
- Filters.

- Map legend.
- Station details.
- Search.

Do not render millions of unfiltered records at once.

Use bounding-box APIs and suitable clustering.

5. MAHARASHTRA ANALYSIS

Implement:
- Maharashtra map.
- District boundaries.
- District selection.
- District comparison table.
- Search and filters.
- EV category breakdown.
- Station distribution.
- District infrastructure indicators.

District table and map selection must remain synchronized.

6. DISTRICT DETAIL

Every supported district detail page must show:
- District geography.
- EV registrations.
- Vehicle category breakdown.
- Charging stations.
- Charging points.
- Infrastructure ratios.
- Spatial coverage.
- Data sources.
- Verification status.

If data is unavailable, explain why.

7. PUNE URBAN GRID

Implement:
- Interactive 1 km grid.
- Coverage and deficit symbology.
- Grid-cell inspector.
- Station markers.
- Grid statistics.
- Layer controls.
- Clear map legend.

Keep grid styling readable.

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ANTIGRAVITY (Agent)

[No response recorded]

Prompt 7 of 12

2026-11-02T18:07:57Z

USER

complete the work properly

ANTIGRAVITY (Agent)

[No response recorded]

Prompt 8 of 12

2026-11-03T04:43:29Z

USER

completethe work proeprluy

ANTIGRAVITY (Agent)

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Prompt 9 of 12 2026-11-03T04:54:24Z

USER

CHARGEATLAS INDIA â€" FINAL PRODUCTION DEPLOYMENT, SECURITY AND COMPETITION ACCEPTANCE

Project directory: D:\EV Infra

Competition: IIT Bombay Techfest 2026â€"27 â€" EV Infrastructure Map India Challenge.

Team: SuperNova, Sanjivani University.

You are completing the final production-readiness stage of an existing application.

The previous development stages have implemented:
- React frontend.
- FastAPI backend.
- PostgreSQL 18.
- PostGIS.
- Authentic data ingestion.
- GIS analytics.
- EV demand forecasting.
- MCDA site recommendations.
- Interactive maps.
- Data management.
- Methodology documentation.

Do not assume these are complete without inspecting the actual code and test evidence.

1. FINAL CODEBASE AUDIT

Inspect the complete repository.

Identify:
- Unused files.
- Duplicate implementations.
- Temporary debugging code.
- Hardcoded development URLs.
- Hardcoded credentials.
- Unused dependencies.
- Broken imports.
- Dead routes.
- Unhandled exceptions.
- Incomplete functions.
- TODO placeholders.
- Fake API responses.

- Synthetic production data references.

Remove or fix only what is safe.

Do not delete useful functionality.

2. SECURITY

Implement or verify:

- Secure password handling.
- Authentication and authorization.
- Protected API routes.
- Secure session or token handling.
- Input validation.
- File-upload validation.
- File-size restrictions.
- File-type restrictions.
- SQL injection protection.
- CORS restrictions.
- Secure HTTP headers where applicable.
- Rate limiting for sensitive endpoints.
- Safe error responses.
- Secret management.
- Logging without passwords or tokens.

Never expose:
- Database credentials.
- API keys.
- Secret tokens.
- Private configuration.
- User passwords.

Review authentication and authorization before production deployment.

3. PRODUCTION CONFIGURATION

Create separate configurations for:

- Local development.
- Testing.
- Production.

Use environment variables for:
- Database URL.
- Database password.
- Secret key.

- Allowed frontend origins.
- External service credentials.
- Map-provider keys, if required.

Create:
.env.example

Do not commit real .env files.

Production must use PostgreSQL/PostGIS.

Production must fail clearly if the production database is unavailable.

Never silently switch production to SQLite.

4. DEPLOYMENT ARCHITECTURE

Evaluate a practical deployment architecture for the existing stack.

Frontend:
- Build the React/Vite application for production.
- Configure static hosting or an appropriate web server.
- Configure client-side route fallback.

Backend:
- Deploy FastAPI using a production ASGI server configuration.
- Configure health checks.
- Configure environment variables.
- Configure logging.
- Configure process restarts.

Database:
- Use a persistent PostgreSQL database with PostGIS support.
- Verify backups.
- Restrict publ

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ANTIGRAVITY (Agent)

[No response recorded]

Prompt 10 of 12 2026-11-03T05:32:01Z

USER

this is the error now

ANTIGRAVITY (Agent)

[No response recorded]

Prompt 11 of 12 2026-11-03T05:38:12Z

USER

https://github.com/Pranav060307/EV-Infra-IIT-bombay-.git PUSH ALL THE work to this git hub repo

ANTIGRAVITY (Agent)

[No response recorded]

Prompt 12 of 12 2026-11-03T06:01:01Z

USER

deploy this on the vercel

ANTIGRAVITY (Agent)

[No response recorded]

ChargeAtlas India — EV Infrastructure Decision Platform
IIT Bombay Techfest 2026-27 | Team SuperNova | Sanjivani University
Lead Developer: Pranav Jadhav
Total prompts in this conversation: 28 | Showing last 12